Web Engineering 1 | WE1

Summary

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1. HYPERTEXT MARKUP LANGUAGE (HTML)

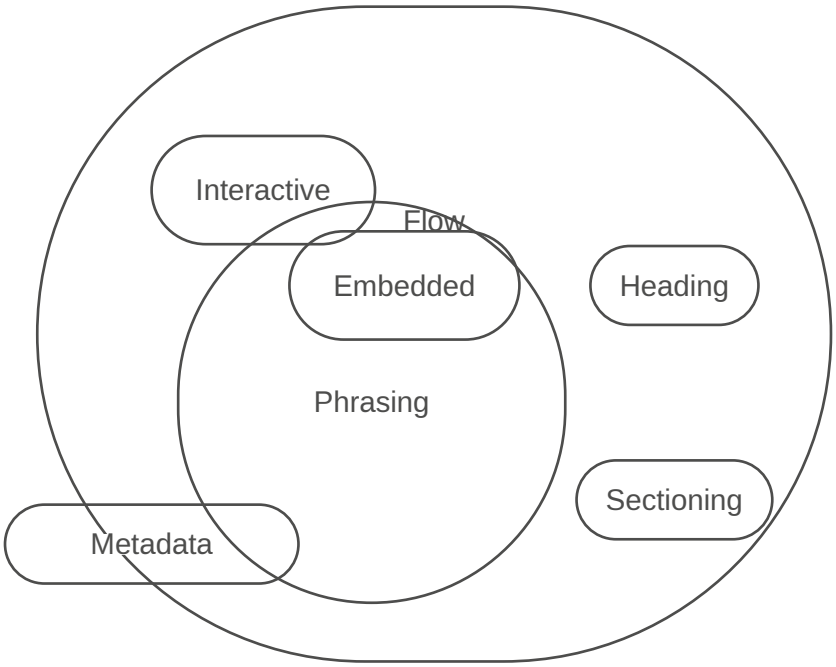
Structures the contents of a document, defines **what** is content and not **how** it looks.

`<head>` contains metadata, `<body>` contains the content.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <link rel="icon" href="ost-favicon-32x32.png">
  <title>Hello World</title>
</head>
<body>
  <h1>Hello World</h1>
  <p>Ich bin eine Testseite</p>
</body>
</html>
```

1.1. CONTENT CATEGORIES

Most HTML elements are a member of one or more content categories. The content categories are used to define the content model of elements, in other words, what each element can take as descendants.



Metadata: Elements belonging to the metadata content category modify the presentation or the behavior of the rest of the document, set up links to other documents, or convey other out-of-band information.

Flow: Flow content is a broad category that encompasses most elements that can go inside the `<body>` element, including heading elements, sectioning elements, phrasing elements, embedding elements, interactive elements, and form-related elements.

Sectioning: Creates a section in the current outline defining the scope of `<header>` and `<footer>` elements.

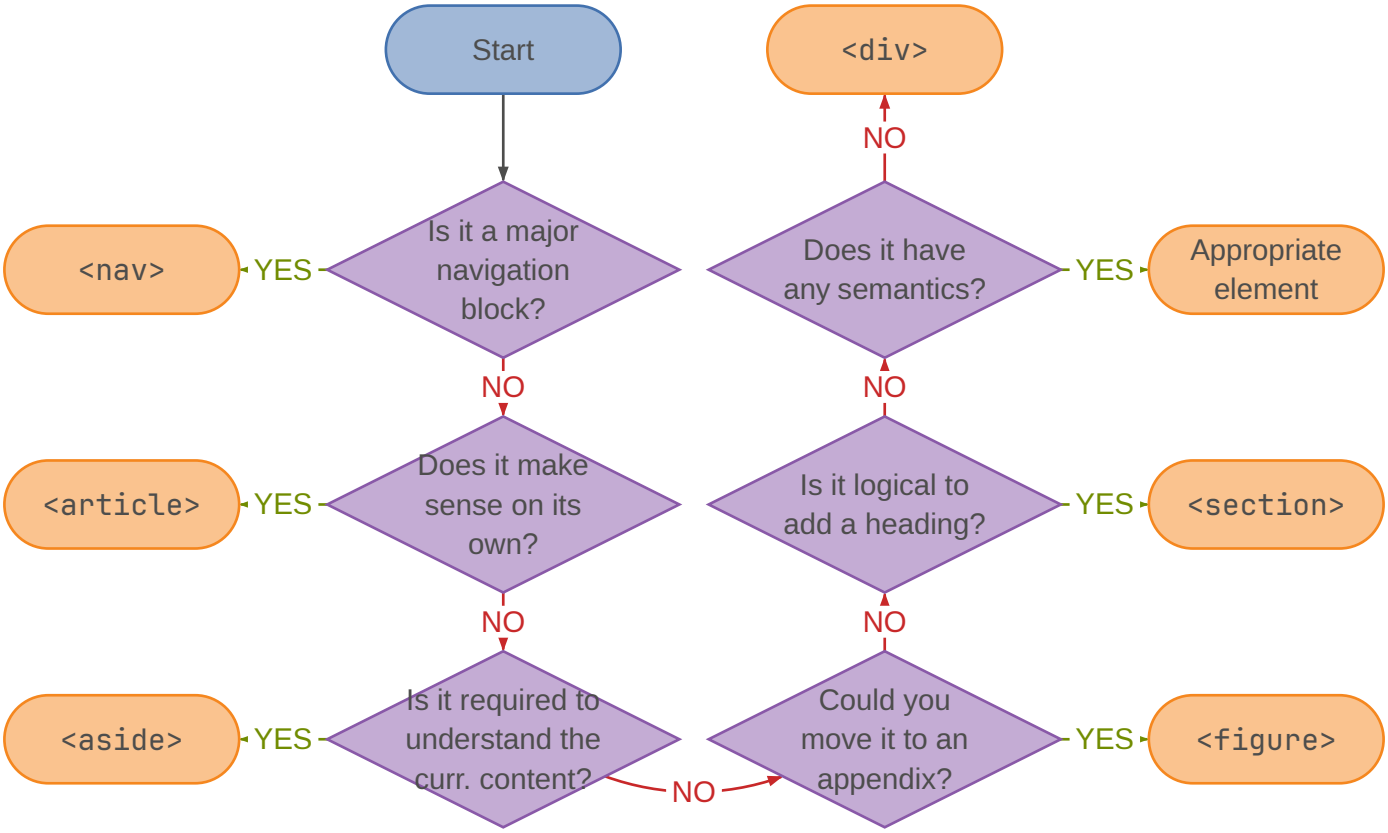
Heading: Defines the title of a section.

Phrasing: Refers to the text and the markup within a document. Sequences of phrasing content make up paragraphs.

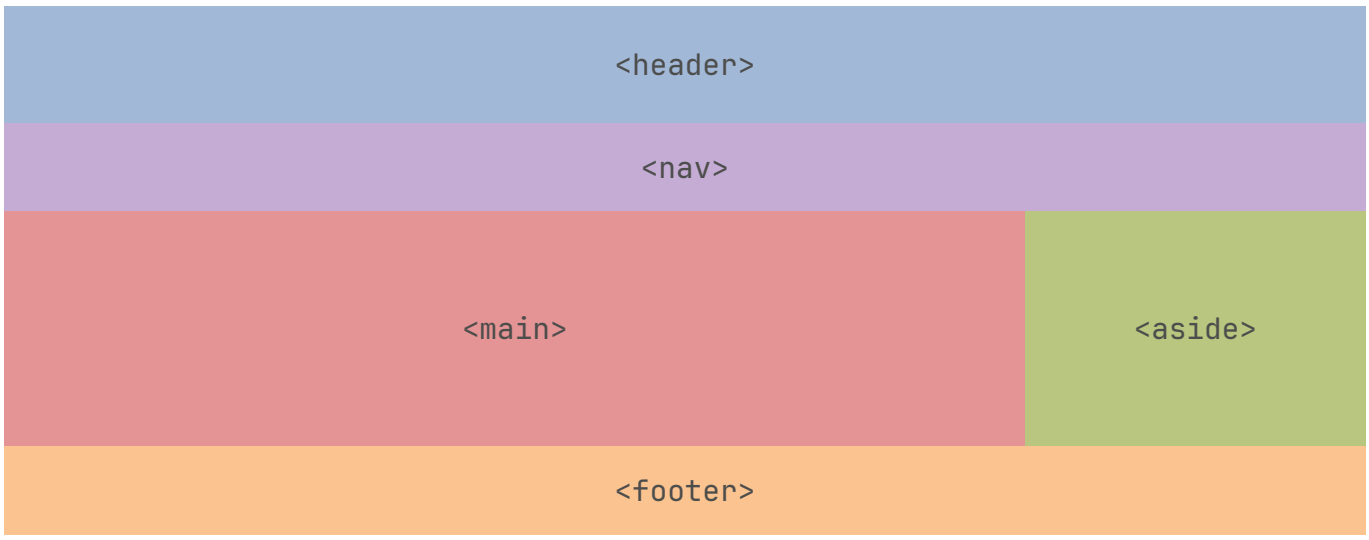
Embedded: Imports another resource or inserts content from another markup language or namespace into the document.

Interactive: Includes elements that are specifically designed for user interaction.

1.2. SEMANTIC ELEMENTS



1.2.1. Webpage structure



1.2.2. h*

<h1> should only appear once on the website.

1.2.3. article

A self-contained part of the document which is independently placed or should be reused.

```
<article>
  <section id="introduction">
    <p>JS was a mistake</p>
  </section>
  <section id="content">
    <!— very long rant —>
  </section>
  <section id="summary">
    <p>JS was a mistake</p>
  </section>
</article>
```

1.2.4. section

General separation and grouping of content.

1.2.5. img

`alt` attribute's value should provide a clear and concise text replacement for the image's content. It should not describe the presence of the image itself or the file name of the image. The `alt` attribute **must be specified**.

If the image is of decorative nature only, one must still set `alt`, although empty: `alt=""`

1.3. VOID ELEMENTS

A void element is an element in HTML that cannot have any child nodes (i.e., nested elements or text nodes). Void elements only have a start tag; end tags must not be specified for void elements.

```
<area> <base> <br> <col> <embed> <hr> <img> <input> <link> <meta> <param>
<source> <track> <wbr>
```

TODO:
https://developer.mozilla.org/en-US/docs/Web/HTML/Reference/Elements

2. CASCADING STYLE SHEETS (CSS)

A simple mechanism for adding style (e.g., fonts, colors, spacing) to Web documents.

Inline: `<p style="color: red;">Text</p>` (Discouraged)

Inside of HTML: `<head><style> p { color: red; } </style></head>` (For small documents)

External file: `<link rel="stylesheet" href="styles.css">` (Standard)

2.1. SELECTORS

Class: `.classname` reusable for multiple elements

ID: `#idname` for unique, single element

Pseudo-class: Prefixed with a colon (`:`), allow the selection of elements based on state information that is not contained in the document tree.

Pseudo-element: Prefixed with two colons (`::`), represent entities that are not included in HTML. For example, the simple `::marker` selector selects list item bullets, and the compound selector `p::first-line` matches the first line of all `<p>` elements.

Universal: The CSS universal selector (*) matches elements of any type.

Nesting: The CSS & nesting selector explicitly states the relationship between parent and child rules when using CSS nesting.

```
parentRule {
  & childRule { }
}
```

Attribute: The CSS attribute selector matches elements based on the element having a given attribute explicitly set, with options for defining an attribute value or substring value match.

```
a[href*="example"] {
  font-size: 2em;
}
```

2.1.1. Combinators

Descendant combinator: The descendant combinator, denoted with one or more spaces, selects nodes that are descendants of the first element.

Child combinator: The child combinator is more specific than the descendant combinator. Denoted with the greater than character (>), the child combinator selects nodes that are direct children of the first element.

Subsequent-sibling combinator: The subsequent-sibling combinator, denoted with a tilde (~), selects siblings. Given `A ~ B`, all elements matching `B` will be selected if they are preceded by `A`, provided both `A` and `B` share the same parent.

Next-sibling combinator: The next-sibling combinator, denoted by the plus symbol (+), is similar to the subsequent-sibling. However, given `A + B`, it only matches `B` if `B` is **immediately** preceded by `A`, with both sharing the same parent.

Column combinator: There is also a column combinator, denoted by two pipe characters (`||`), which, when supported, selects nodes that belong to a column.

Namespace separator: The namespace separator is another combinator that is generally used in conjunction with the `@namespace` at-rule. This combinator is denoted by a single pipe character (`|`). It enables limiting type selectors and the universal selector to a specific namespace.

2.1.2. Specificity

1) `!important`
2) `style=""` Attribute
3) (1) ID-Selectors
4) (2) Class-Selectors, Pseudoclasses (eg. `:hover`) and Attributes (`[...=...]`)
5) (3) Type-Selectors and Pseudoelements (eg. `::first-line`)

> ***Example** 1:*

```
div.help > div#nested a:hover::before { }
```

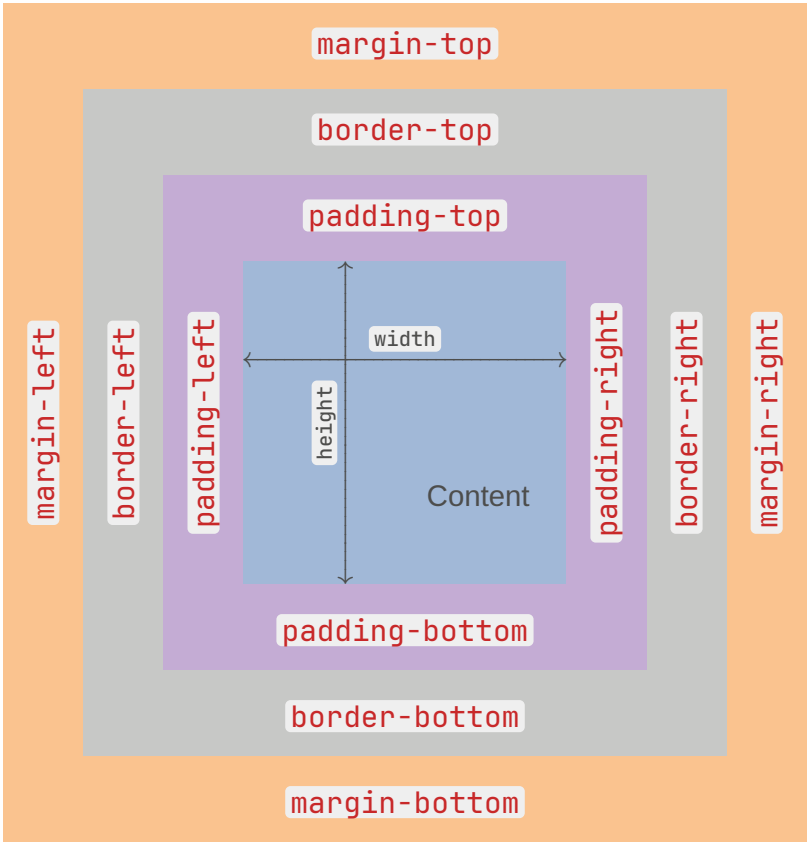
Specificity: 124
1 id, 2 (pseudo-)classes, 4 (pseudo-)elements)

***Example** 2:*

```
<div id="div-1" class="recent significant good-to-go">1</div>
```

```
.recent // overrides others
```

2.2. BOX-MODEL



2.3. DISPLAY

Term	*Definition*
Block (`p, h1, ul`)	Vertical alignment, width fills parent, height from content
Inline (`a, strong`)	Horizontal alignment, width and height from content, `left/right margin/padding` allowed, not `top/bottom`
Inline-Block	Flows like inline, but `width/height margin/padding` all allowed

`display: none;` Element isn't visible and doesn't take any space `visibility: hidden;` Element isn't visible but does use space

2.3.1. Flexbox

flexbox.help

3. JAVASCRIPT (JS)

why did you do this to us, brendan eich

3.1. ATOMIC LANGUAGE CONSTRUCTS

3.1.1. Values and types

JS is dynamically typed (help) and doesn't have constant values. It only has non-reassigneable name-spaces when using `const`. Using `let` you declare a reassigneable variable and using `var` is a crime against humanity.

String interpolation:

```
let answer = 42;
console.log(`The answer is ${answer}`);
```

3.1.2. Functions

Functions are hoisted to the top. A function wihout an explicit return returns `undefined`.

```
function printName(name) {
  console.log(name);
}
```

Lambda functions

```
((a, b) ⇒ console.log(a + b == 5))(2, 3)
```

3.1.3. Objects

```
let obj = {
  key1: 69,
  key2: "nice",
};

let arr = [ obj, 1, "cursed" ];
```

3.1.4. Equality

Equality in js is weird. Double equals (==) will perform a type conversion when comparing two things, and will handle `NaN`, `-0`, and `+0` specially to conform to IEEE 754 (so `NaN ≠ NaN`, and `-0 == +0`). Triple equals (===) will do the same comparison as double equals (including the special handling for `NaN`, `-0`, and `+0`) but without type conversion; if the types differ, `false` is returned.

4. DOCUMENT OBJECT MODEL (DOM)